

Python Star Pattern Assignment

Medium to Hard Level — 2 Questions

Instructions

Solve both questions using Python. Use nested loops and conditional statements. The program should work for different values of **n**. Do not hard-code the pattern.

Question 1 — Hollow Diamond

Write a Python program that takes an **odd number n** from the user and prints a hollow diamond pattern.

Example Input: n = 7

Expected Output:

```
  *
 * *
*   *
*   *
*   *
 * *
  *
```

Requirement: Use nested loops. Do not create the upper and lower parts by manually writing separate print statements.

Hint: Think about the number of spaces before the first star and the number of spaces between the two stars.

Question 2 — Butterfly Pattern

Write a Python program that takes the number of rows **n** from the user and prints a butterfly star pattern.

Example Input: n = 5

Expected Output:

```
  *      *
 **     **
***    ***
****   ****
***** *****
****   ****
***    ***
**     **
 *      *
```

Requirement: Use nested loops to control the stars and spaces. Your program should work for any positive value of **n**.

Hint: Divide the pattern into an increasing section and a decreasing section. For each row, calculate the number of stars on both sides and the spaces in the middle.

Submission Requirements

- Write clean and properly indented Python code.
- Use meaningful variable names.
- Test both programs with at least two different values of n .
- Do not copy the solution from the internet.